Approach-Region Dynamics and Rate Turnover in Thermally Activated Barrier Crossing

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Abstract

Chemical reaction dynamics has a well-developed thermodynamic description of barrier crossing, but a less operational account of the dynamical conditions governing whether a reactive trajectory commits productively or recrosses. This work shows that as a reactive coordinate approaches a transition state through a stable precursor basin, this behavior is organized by a dimensionless damping ratio, $\chi_0 = \Gamma_{\text{eff}}/(2\Omega_0)$, where Ω_0 is the precursor-well frequency and Γ_{eff} is the total projected dissipation. Below $\chi_0 = 1$ approach dynamics are oscillatory and recrossing-prone; above it monotonic but increasingly sluggish; at $\chi_0 = 1$ the approach coordinate reaches an exceptional point and monotonic commitment is fastest. Established descriptions of barrier crossing in electron transfer, proton-coupled electron transfer, and heterogeneous surface reactivity yield locally equivalent precursor-well damped-generator dynamics despite distinct microscopic origins. Three calibrated estimator routes connect χ_0 to spectroscopic linewidths, solvent longitudinal relaxation times, and electronic friction coefficients, each with explicit validity bounds. Worked examples illustrate solvent-dependent turnover in TEMPO electron transfer and promoter-dependent N_2 activation on Fe surfaces. The approach-dynamics descriptor complements thermodynamic rate theory by quantifying when barrier crossing is limited by recrossing, overdamped sluggishness, or near-critical commitment.

Keywords: damping ratio; barrier crossing; electron transfer; reaction dynamics; heterogeneous catalysis; Kramers turnover; PCET

Introduction

Rate theory has a well-understood thermodynamic axis: transition-state theory connects barrier height and driving force to rate constants, and Brønsted–Evans–Polanyi (BEP) scaling relations organize these across reaction families. What thermodynamic theory does not address is whether the reactive coordinate, having arrived at the transition state, actually completes the transformation or recrosses back. This gap is experimentally real. Identical donor–acceptor pairs with the same ΔG° and $\Delta G^\ddagger$ produce dramatically different charge-separated-state populations across solvents. PCET systems shift between stepwise and concerted mechanisms under changes in environmental control parameters such as solvent, composition, or pressure, even when the thermodynamic landscape remains broadly similar. Fe(111) outperforms Fe(110) in N_2 activation despite comparable binding energies. In each case, the missing variable is dynamical commitment. The present work is therefore framed as a reaction-dynamical extension to established barrier-crossing theory, aimed at identifying a physically measurable control parameter for commitment versus recrossing across chemically distinct systems.

The reaction-dynamical ratio $\chi = \Gamma_{\text{eff}}/(2\Omega_0)$, defined as the total projected dissipation rate divided by twice the precursor-well approach frequency, was introduced in prior work [29] as the parameter governing solution character near the transition state: below $\chi = 1$ the approach motion is oscillatory and recrossing-prone, above it monotonic but increasingly sluggish, and at $\chi = 1$ productive commitment is fastest. The present paper operationalizes this quantity through three calibrated estimator routes connecting χ to standard laboratory and computational observables, with interpretation protocols, worked examples, and explicit validity boundaries, making χ a routinely computable and falsifiable descriptor alongside thermodynamic rate theory.

Section 1 establishes the generator and domain mapping. Section 2 derives the three estimator routes. Section 3 provides the quantitative interpretation protocol. Section 4 develops the design levers with domain-specific worked examples and experimental predictions. Section 5 defines the validity boundaries. Derivations, bias analysis, and the complete falsification protocol are in the Supplementary Information.

Methodological Refinements. The framework described here requires several structural departures from phenomenological treatments of reaction-coordinate damping. Table 1 documents these explicitly, distinguishing the rigorous topological formulation from approximations that appear in earlier treatments and in prior drafts of this work, and shifting the analytical focus from saddle-point heuristics to the stable precursor-well approach dynamics.

Framework Element	**Prior formulation**	**Corrected formulation**
Generator topology	$\chi = 1$ EP2 mapped to the saddle-point transition state	EP2 belongs exclusively to the stable precursor-well approach dynamics; the inverted barrier has no discriminant crossing
Route B scaling	Damping $\Gamma = 1/\tau_L$ (inverse relaxation rate); $\chi \propto 1/\tau_L$	Damping $\Gamma = C_{\text{solv}}\tau_L$ (Debye-Drude friction); $\chi \propto \tau_L$; slower solvents are strictly more overdamped
PCET isotope shift	H$\to$D lowers Ω and lowers χ: $\chi^D_{\text{eff}} \approx 0.71\,\chi^H_{\text{eff}}$	H$\to$D lowers Ω_0 and raises χ: $\chi^D_{\text{eff}} = \sqrt{2}\,\chi^H_{\text{eff}} \approx 1.41\,\chi^H_{\text{eff}}$ (upper bound; proton-dominated limit)
Route C dimensionality	Cross-check linewidth: $\Delta\tilde{\nu} = \hbar\Gamma_{NN}/(hc)$	Γ_{NN} is already an energy width (eV); correct cross-check: $\Delta\tilde{\nu} = \Gamma_{NN}/(hc)$; no additional $\hbar$ factor

Table 1: Structural corrections to prior formulations. These refinements retain the central approach-dynamics damping descriptor χ_0 while grounding its physical interpretation in the stable precursor-well coordinate and enforcing dimensional consistency across all three estimator routes.

1 The Dynamical Ratio: Generator, Boundary, and Domain Mapping

Near any transition state, projection of the full potential energy surface onto the committor-dominant coordinate reduces locally to a second-order damped oscillator in the *precursor well*:

$$\ddot{q} + \Gamma_{\text{eff}}\,\dot{q} + \Omega_0^2\,q = \xi(t) \tag{1}$$

where Ω_0 is the characteristic frequency of the stable approach coordinate in the precursor well ($\Omega_0^2 > 0$) and Γ_{eff} is the total projected dissipation from all channels: solvent friction, vibrational coupling, or electronic hybridization with a metal surface, depending on the domain.

The reactive coordinate at the saddle point itself has inverted curvature, $\ddot{q} + \Gamma_{\text{eff}}\dot{q} - \omega_b^2 q = 0$, with characteristic roots $\lambda_\pm = -\Gamma_{\text{eff}}/2 \pm \sqrt{\Gamma_{\text{eff}}^2/4 + \omega_b^2}$, whose discriminant $\Gamma_{\text{eff}}^2 + 4\omega_b^2 > 0$ for all Γ_{eff}, $\omega_b > 0$. No exceptional point exists at the saddle. The EP2 structure belongs entirely to the precursor-well generator Eq. (1): the discriminant $\Gamma_{\text{eff}}^2 - 4\Omega_0^2$ changes sign at exactly one point,

$$\chi_0 \;\equiv\; \frac{\Gamma_{\text{eff}}}{2\Omega_0} \;=\; 1 \tag{2}$$

This is the approach-dynamics boundary. The notation χ is used throughout for χ_0 when the approach-region context is clear. The transition-state value χ_{TS}, defined using saddle-point geometry, is treated separately as a post-commitment transit diagnostic: it is not the EP variable, does not set the turnover maximum, and large χ_{TS} slows rather than accelerates reactive descent (Supplementary §3.3). The approach-dynamics χ_0 is the quantity that governs whether trajectories commit or recross, and is what all three estimator routes compute.

Below this boundary ($\chi < 1$) solutions are oscillatory and recrossing occurs. Above it ($\chi > 1$) solutions decay monotonically but with increasing sluggishness. At $\chi = 1$, monotonic commitment is fastest; the generator reaches a second-order exceptional point [1] where eigenvalues coalesce and the Green's function becomes polynomial-exponential (Supplementary §1.1). This is a mathematical property of the generator Eq. (1), not a chemical model. Its relevance here is specifically to chemical barrier crossing: the same local generator organizes reactive commitment in multiple classes of chemically distinct systems.

Three independently derived theoretical frameworks (Kramers (1940) [2] (reviewed comprehensively in Hänggi et al. 1990 [3]), Zusman (1980) [4], and Newns–Anderson (1969) [5]) yield linearized barrier-crossing dynamics that are locally equivalent to this generator when projected onto the committor-dominant coordinate, as summarized in Table 2. χ is the single ratio that organizes all three.[1]

[1] The physical origin of Γ_{eff} differs across domains: nuclear Langevin friction in Kramers, dielectric solvent relaxation in Zusman, electronic hybridization width in Newns–Anderson. The mathematical structure of the generator is identical. The claim is structural reduction, not ontological identity.

Domain	Γ_{eff} **is**	Ω_0 **(approach frequency) is**	**Primary dynamical role of** χ
ET / Kramers	Solvent friction $C_{\text{solv}}\tau_L$ or reorganization damping	Solvation coordinate frequency $\sqrt{2\lambda/(m_{\text{eff}}\Delta Q^2)}$	Friction-controlled recrossing vs. commitment; interacts with electronic adiabaticity (Zusman g)
PCET	Coupled solvent and vibronic damping Γ_{eff}	Frequency of the combined PT–ET precursor-well coordinate	Stepwise vs. concerted mechanism selection
Catalysis (Newns–Anderson)	Electronic friction from d-band hybridization Γ_{NN}	Adsorbate stretch frequency at ground-state geometry, softened by π^*-backdonation	Bond activation efficiency; BEP outliers

Table 2: Domain instantiations of the generator Eq. (1). The physical origin of Γ_{eff} differs across domains; the mathematical structure is identical.

The reduction is reaction-dynamical: each framework's linearized committor-projected barrier-crossing dynamics is governed by the same second-order operator despite differing microscopic origins. No assumptions of shared ontology are made.

The three dynamical regimes and their chemical consequences are given in Table 3. Two points about these regimes deserve emphasis. First, the dynamical optimum is a window $\chi \in [0.8, 1.3]$, not the point $\chi = 1.000$ exactly. The window is derived from the criterion $\eta \geq 0.95$ applied to the symmetric schematic efficiency curve $\eta(\chi) = 2\chi/(1+\chi^2)$: $\eta(0.8) = 0.976$ and $\eta(1.3) = 0.967$, both satisfying the criterion independently of any dataset. The Pollak–Grabert–Hänggi (PGH) combined rate [6] provides a complementary conservative estimate of $[0.8, 1.2]$ at $V^\ddagger/k_BT \geq 8$; the operational window $[0.8, 1.3]$ lies between this estimate and the schematic 95% band $[0.72, 1.38]$ (Supplementary §1.2, with explicit numerical verification). Second, Route C ground-state estimates provide lower-bound proxies for the approach-region χ_0 in systems where the reactive bond softens toward the saddle: bond softening decreases Ω_0 while Γ_{eff} stays constant or increases, raising χ along the reaction path. A ground-state Route C diagnosis of $\chi_{\text{GS}} \approx 0.6$ is therefore consistent with $\chi_0 \approx 1$ at the approach region if the bond softens significantly. Routes A and B estimate the approach-region χ_0 directly from spectroscopic or solvent-dynamics quantities and should not be interpreted as lower bounds on a saddle-point χ_{TS}; in ET and PCET systems the solvent coordinate and coupled proton–electron coordinate do not follow the same bond-softening path as a surface adsorbate.

χ	Regime	Coordinate behavior	Chemical consequence
< 0.8	Underdamped	Oscillatory; recrossing probable	Back-transfer; poor yield; weak bond activation
0.8–1.3	Near-critical	Fastest monotonic commitment	Maximum ET yield; concerted PCET; peak catalytic TOF
> 1.2	Overdamped	Monotonic but sluggish	Solvent-controlled ET; inhibited PCET; poisoned catalyst

Table 3: The three dynamical regimes. The window $[0.8, 1.3]$ captures $\geq 95\%$ of the available dynamical efficiency under physically relevant barrier heights (Supplementary §1.2).

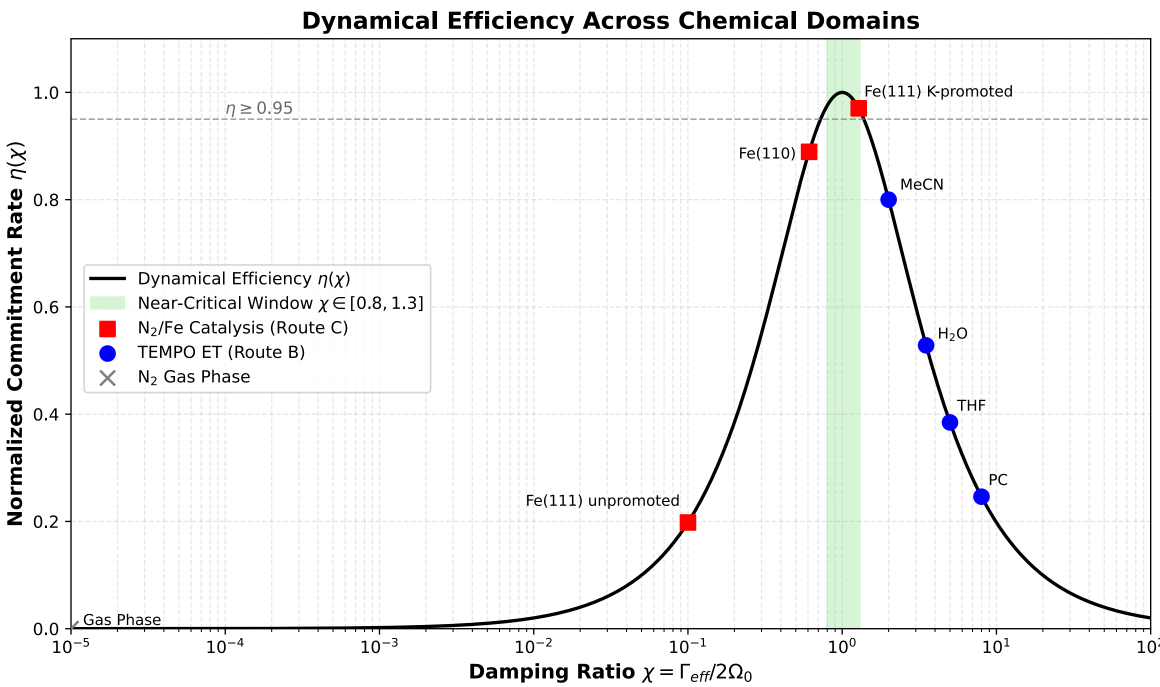

Figure 1: Dynamical efficiency across domains. Reduced symmetric turnover schematic $\eta(\chi) = 2\chi/(1 + \chi^2)$ as a function of the approach-dynamics damping ratio $\chi = \Gamma_{\text{eff}}/(2\Omega_0)$, plotted on a log scale. This curve visualizes the energy-diffusion/spatial-diffusion turnover structure; it is not a full numerical PGH/Mel'nikov–Meshkov rate calculation and not an independent empirical fit. Dotted vertical lines: exact $\eta \geq 0.95$ roots of the schematic curve, $\chi \in [0.72, 1.38]$; dashed vertical lines: operational near-critical band $\chi \in [0.8, 1.3]$. Main-panel markers (red squares) are Route C absolute estimates for N_2/Fe cases; their y-values are model projections onto $\eta(\chi)$, not measured efficiencies. **Inset:** TEMPO Route B relative χ series from τ_L only, normalized to acetonitrile; this is not an absolute χ placement, and quantitative cross-class solvent comparison requires independent solvent librational frequencies Ω_s.

The framework applies only when the transition-state region is locally harmonic in the reactive coordinate. Strongly anharmonic transition states invalidate the generator Eq. (1); the committor-projected potential should be verified to be parabolic before applying χ. For surface adsorbates in highly activated dissociation (e.g. N_2 on Fe), the precursor well is anharmonic: Morse-type bond softening along the approach path progressively lowers Ω_0 below the ground-state harmonic value. Route C ground-state estimates therefore underestimate χ_0 along the approach coordinate; the true approach-region damping ratio is higher than the harmonic estimate, consistent with the lower-bound interpretation noted in Section 2. When multiple vibrational modes couple comparably to the reactive coordinate, the scalar χ must be replaced by the mode-projected effective quantity χ_{eff} (Supplementary §6.1); a single χ is adequate only when one mode carries more than 70% of the reactive weight.

2 Estimator Routes: Three Paths from Observables to χ

For theoretical reaction dynamics to be chemically useful, the damping ratio must be inferable from observables or from standard electronic-structure-based quantities rather than treated as a purely formal parameter. Three estimator routes serve this purpose, each with a defined domain of validity, a defined bias, and explicit conditions for inapplicability. Because χ_0 describes the approach region rather than the saddle itself, direct saddle-point measurements are not required; estimators yield χ_0 or ground-state proxies. Routes should not be mixed within a single comparison.

Route	**Inputs**	**Formula**	**Applies when**	**Bias and limit**
A: Spectroscopic	$\Delta\tilde{\nu}_L$, $\tilde{\nu}$ (cm^{-1})	$\chi_{\text{spec}} = \Delta\tilde{\nu}_L/(2\tilde{\nu})$	Operando IR or Raman; surface catalysis	Valid only when Lorentzian fraction $> 50\%$; linewidth must be coverage-independent; cross-check with T_1 if available
B: Solvent relaxation	τ_L (ps), λ (eV), ΔQ (Å), m_{eff} (amu)	$\Omega_0 = \sqrt{2\lambda/(m_{\text{eff}}\Delta Q^2)}$; $\chi_{\text{ET}} = C_{\text{solv}}\tau_L/(2\Omega_0)$, $C_{\text{solv}}=\Omega_s^2$ (solvent libration); relative form $\chi_{\text{ET}}^{(A)}/\chi_{\text{ET}}^{(B)} = \tau_L^{(A)}/\tau_L^{(B)}$ preferred	Electron transfer; PCET; solvent-controlled kinetics	C_{solv} and m_{eff} uncertain; relative χ eliminates m_{eff}, λ, ΔQ exactly; C_{solv} cancels only for same-class solvents
C: DFT friction	Γ_{NN} (eV), $\tilde{\nu}$ (cm^{-1})	$\chi_{\text{DFT}} = (\Gamma_{\text{int}} + \Gamma_{NN})/(2hc\tilde{\nu})$; $hc = 1.240 \times 10^{-4}$ eV·cm	Surface catalysis with DFT electronic friction available	Mean-field DFT must be valid; 3d ferromagnets may require DFT+U correction

Table 4: Three estimator routes for the approach-region damping ratio χ_0. Route C ground-state values serve as lower-bound proxies for χ_{TS} only when the reactive bond softens along the reaction path.

Route A: Spectroscopic estimator

Route A uses vibrational linewidth as a proxy for the dissipation rate. The true dissipation rate is $\Gamma = 1/T_1$ (vibrational population lifetime); the measured IR linewidth reflects T_2 coherence decay via $1/T_2 = 1/(2T_1) + 1/T_2^*$, where T_2^* encodes pure dephasing and inhomogeneous broadening. Route A gives $\chi_{\text{spec}} = \Delta\tilde{\nu}_L/(2\tilde{\nu})$, which equals the true damping ratio of the measured vibrational mode only when $T_2 \approx 2T_1$, i.e., when homogeneous lifetime broadening dominates. In precursor-state spectroscopy this is an approach-region estimate; it should not be interpreted as the saddle-point χ_{TS} unless the measured mode is independently confirmed to correspond to the saddle-region normal mode.

The raw FWHM from spectrometer software must not be used. Every peak must be fit to a Voigt profile, and only the Lorentzian component $\Delta\tilde{\nu}_L$ extracted. When the Gaussian component exceeds the Lorentzian component, the sample is either too warm or too inhomogeneous, and the linewidth no longer reflects the dissipation channel; the estimate will be spuriously high, and the measurement should be repeated at lower temperature or lower adsorbate coverage. The measurement procedure is: (1) acquire operando IR under steady-state turnover conditions; (2) fit a Voigt profile and extract $\Delta\tilde{\nu}_L$ separately from $\Delta\tilde{\nu}_G$; (3) confirm Lorentzian fraction $> 50\%$ and coverage-independence of the linewidth; (4) compute $\chi_{\text{spec}} = \Delta\tilde{\nu}_L/(2\tilde{\nu}_{\text{peak}})$; (5) if T_1 data are available from pump–probe vibrational spectroscopy, verify that $\Gamma_{\text{true}} = 1/T_1$ gives consistent χ within a factor of two. Lorentzian lineshape is necessary but not sufficient proof of homogeneous broadening; coverage independence and T_1 cross-checks are essential.

On metallic substrates, interference between a discrete molecular vibration and a continuous electronic or phonon background can produce Fano asymmetry in the observed lineshape, rendering Voigt decomposition unreliable for extracting the homogeneous linewidth. In such cases Route A should be extended to a direct Fano fit,

$$I(E) \propto \frac{(q+\epsilon)^2}{1+\epsilon^2}, \qquad \epsilon = \frac{2(E-E_R)}{\Gamma}, \tag{3}$$

which explicitly extracts the intrinsic homogeneous linewidth Γ without requiring a Lorentzian component. The asymmetry parameter q and the linewidth Γ are independent quantities; large $|q|$ (symmetric profile) does not imply small Γ, and small $|q|$ (window resonance) does not imply large Γ. Route A remains applicable whenever the Fano fit converges reliably, regardless of lineshape asymmetry.

Route C (Γ_{NN} from DFT electronic friction) may be used when no experimental spectrum is available, but should be treated as a *lower bound* on χ_0: it captures electronic friction only and systematically underestimates the total dissipation when phononic or bath friction contributes significantly to the approach-region dynamics. Routes A and C are complementary, not mutually exclusive; when both are available, consistency between them confirms that electronic friction dominates, while $\chi_{\text{spec}} > \chi_{\text{DFT}}$ signals a non-negligible phononic contribution.

Route B: Solvent-relaxation estimator

Route B connects χ to the longitudinal dielectric relaxation time τ_L. In a Debye-Drude polar solvent, the friction coefficient on the solvation coordinate scales as $\gamma_{\text{solv}} = m_{\text{eff}}\,\Omega_s^2\,\tau_L$, where Ω_s is the characteristic solvent librational frequency (typically 50–150 cm^{-1} for common polar solvents) [3]. The damping rate on the precursor-well coordinate is therefore $\Gamma_{\text{eff}} = \gamma_{\text{solv}}/m_{\text{eff}} = \Omega_s^2\,\tau_L \equiv C_{\text{solv}}\,\tau_L$, so $\chi \propto \tau_L$: slower, more viscous solvents exert greater drag and produce larger χ, consistent with the lever assignments in Table 6. Because C_{solv} and m_{eff} are not independently accessible from ET kinetics, an absolute Route B estimate carries order-of-magnitude uncertainty and should not be reported to more than one significant figure.

The operationally robust quantity is the relative χ. For a fixed donor–acceptor pair across a solvent series, this formulation strictly assumes isoenergetic reorganization ($\lambda^{(A)} \approx \lambda^{(B)}$). Provided the variation in the longitudinal relaxation time τ_L dominates any solvent-dependent fluctuations in the Pekar factor, the parameters m_{eff} and ΔQ cancel exactly, and $C_{\text{solv}} = \Omega_s^2$ cancels to the extent that Ω_s is comparable across the series:

$$\frac{\chi_{\rm ET}^{(A)}}{\chi_{\rm ET}^{(B)}} \approx \frac{\tau_L^{(A)}}{\tau_L^{(B)}} \tag{4}$$

where equality holds in the limit $\Omega_s^{(A)} = \Omega_s^{(B)}$. For cross-class comparisons (e.g. water vs. THF), Ω_s can differ by a factor of two to three, and the corrected form $\chi_{\rm ET}^{(A)}/\chi_{\rm ET}^{(B)} = (\Omega_{s,A}^2/\Omega_{s,B}^2) \times (\tau_L^{(A)}/\tau_L^{(B)})$ should be used.

This is the operationally robust quantity. To locate the dynamical optimum, compile $k_{\rm et}$ and τ_L across at least five solvents and plot $k_{\rm et}$ against τ_L. The rate maximum identifies the solvent where $\chi_{\rm ET} \approx 1$. Solvents with shorter τ_L are less damped (underdamped); solvents with longer τ_L are more damped (overdamped). Eq. (4) maps the entire series relative to the turnover without requiring any absolute parameter. To confirm solvent-dynamical control, verify that $k_{\rm et} \times \tau_L$ is approximately constant across the series. If a bracketed absolute estimate is required, compute $\chi_{\rm ET}$ over a range of $m_{\rm eff} \in \{50, 100, 200, 500\}$ amu and $C_{\rm solv}$ values and report the full bracket; a regime assignment is reliable only if the entire bracket falls within a single regime.

Route B is not intended to supersede established solvent-dynamical descriptors such as τ_L or dielectric friction coefficients, but to place them in competition with the intrinsic precursor-well stiffness (Ω_0) to identify the dimensionless critical boundary ($\chi_0 = 1$) shared across all three reaction classes considered here. The longitudinal relaxation time alone cannot locate this boundary because it carries no information about Ω_0; the dimensionless ratio χ_0 provides the missing comparison.

Route C: DFT electronic friction estimator

Route C is the highest-fidelity estimator when electronic friction dominates, as in chemisorption on metal surfaces. The mode-projected electronic friction tensor Γ_{NN} is obtained from density-functional perturbation theory [13]. Here $\Gamma_{\rm cat} \equiv \Gamma_{NN}$ denotes the Newns–Anderson hybridization energy broadening $2\pi|V_k|^2\rho(\varepsilon_F)$ in eV, which is an electronic energy width, not a vibrational friction coefficient. The vibrational energy $E_{\rm vib} = hc\,\tilde{\nu}_{\rm bond}$ is also in eV, so the estimator:

$$\chi_{\rm DFT} = \frac{\Gamma_{\rm int} + \Gamma_{\rm cat}}{2\,hc\,\tilde{\nu}_{\rm bond}} \qquad hc = 1.23984 \times 10^{-4}\ \text{eV}\cdot\text{cm}$$

is the ratio of two energies and is therefore strictly dimensionless.

As a qualitative consistency cross-check, the spectroscopic linewidth corresponding to $\Gamma_{\rm cat}$ is $\Delta\tilde{\nu} = \Gamma_{\rm cat}/(hc)$ in cm^{-1} (no additional $\hbar$ factor needed). This electronic hybridization linewidth probes a different dissipation channel than the vibrational T_1/T_2 lifetime measured by Route A EELS/IRAS; exact numerical agreement is not expected. Order-of-magnitude consistency with $\Delta\tilde{\nu}_L$ from Route A confirms that the electronic and vibrational friction scales are not grossly discrepant, which is the appropriate cross-check criterion for these distinct observables. Route C computes χ at the ground-state adsorbate geometry and is therefore a lower bound on $\chi_{\rm TS}$: as the reactive bond stretches toward the transition state, π^*-backdonation increases (raising $\Gamma_{\rm cat}$) while $\Omega_{\rm bond}$ decreases, both effects raising χ above the ground-state value. For quantitative transition-state predictions, Γ_{NN} and the harmonic frequency should be computed at the saddle-point geometry along the reaction path. For screening purposes, $\chi_{\rm DFT}$ from the ground-state geometry is adequate provided it is interpreted as a lower bound.

3 Quantitative Interpretation and the Regime Decision Protocol

Once a χ estimate is in hand, interpretation requires first verifying that thermodynamic parameters (ΔG°, λ, $\Delta G^\ddagger$) are independently acceptable. χ addresses only the dynamical axis and cannot compensate for an unfavorable barrier or driving force. With thermodynamics confirmed, the regime assignment in Table 5 guides the appropriate response.

χ	**Diagnosis**	**Thermodynamics check**	**Recommended response**
< 0.5	Strongly oscillatory; recrossing dominates	Thermodynamic parameters may be fine; bottleneck is dynamical	Increase Γ_{eff} substantially; confirm system is in scope
0.5–0.8	Oscillatory; commitment partial; back-transfer significant	Verify ΔG°; if favorable, dynamical adjustment is worthwhile	Moderate increase of Γ_{eff} or decrease of Ω_0; targeted solvent or promoter change
0.8–1.3	Dynamically optimized	Thermodynamics is now the primary constraint	Do not adjust dynamics; optimize barrier and driving force instead
1.3–2.0	Monotonic but slowing; solvent-controlled or promoter-poisoned	Thermodynamic driving force may be partially consumed by excess friction	Reduce Γ_{eff} or increase Ω_0; faster solvent; reduce promoter loading
> 2.0	Dynamics severely overdamped; coordinate sluggish	Check for thermodynamic over-binding; may require new scaffold	Major redesign: decouple bath modes, change metal or facet

Table 5: Regime interpretation protocol. The diagnosis assumes thermodynamic parameters have been independently verified.

An underdamped system is not always performing sub-optimally. If the electron-transfer rate is already faster than the downstream process, adjusting χ adds no value. The regime assignment identifies the source of limitation, not a universal performance ranking. The appropriate action in each case depends on whether the dynamical axis is actually the binding constraint.

4 Design Levers, Worked Examples, and Experimental Predictions

The optimization objective is to tune Γ_{eff} and Ω_0 such that $\chi_0 \approx 1$ while holding thermodynamic parameters fixed. Changing ΔG°, λ, or $\Delta G^\ddagger$ simultaneously with dynamical variables invalidates a χ-based diagnosis and obscures which axis was responsible for any observed change. Two independent handles exist in every domain: Γ_{eff} can be raised or lowered by modifying the coupling between the reactive coordinate and its bath, and Ω_0 can be raised or lowered by modifying the stiffness of the precursor-well coordinate. In heterogeneous catalysis, a third handle modulates how the reactive coordinate projects onto the friction tensor through mode geometry and surface coordination.

Table 6 summarizes the domain-specific lever assignments. Worked examples follow for each domain.

Domain	**Raise** χ	**Lower** χ	**Key observable**
Electron transfer	More viscous or slower solvent ($\uparrow \tau_L$); larger λ or ΔQ; heavier donor–acceptor pair	Faster-relaxing solvent ($\downarrow \tau_L$); small-λ structural redesign; less polar medium	$\log k_{\mathrm{et}}$ vs. $\log \tau_L$; slope ≈ -1 (high-friction branch) or $k_{\mathrm{et}} \times \tau_L \approx$ const. indicates dynamical control
PCET	Increase solvent viscosity; hydrostatic pressure; strengthen H-bond network; H $\rightarrow$ D substitution raises χ by $\times\sqrt{2}$	Decrease viscosity; reduce pressure; disrupt H-bond donors	KIE and pH-independence as a function of viscosity or pressure
Heterogeneous catalysis	Promoter loading (K, Cs); alloy with more oxophilic metal to increase d-band coupling and electronic friction; π^*-backdonation lowers $\tilde{\nu}$	Reduce promoter; alloy with noble metal to weaken d-band hybridization; increase bond stiffness or partially oxidize surface	Operando $\Delta\tilde{\nu}_L/\tilde{\nu}$ vs. TOF; maximum at $\chi \approx 1$

Table 6: Domain-specific tuning levers. All adjustments should hold ΔG°, λ, and $\Delta G^\ddagger$ as fixed as possible.

Electron transfer: optimizing solvent for maximal yield via relative χ

The following worked example is literature-consistent and is intended as an illustration of the reaction-dynamical interpretation of Route B, rather than as a claim of a global fit to a single fully controlled experimental solvent series.

A donor–acceptor pair with $\lambda = 0.7$ eV and $\Delta G^\circ = -0.3$ eV shows incomplete charge separation across five solvents despite favorable thermodynamics (illustrative of the phenomenology described in Barbara et al. [8], Marcus [24], Grampp and Rasmussen [25], and Kumpulainen et al. [9]). The measured rate constants and longitudinal relaxation times are given in Table 7.

Solvent	τ_L (ps)	k_{et} (10^8 M^{-1}s^{-1})	χ_{rel} (τ_L-only)	χ_{rel} (corrected^a)
Acetonitrile	0.20	5.0	1.0 (ref)	1.0 (ref)
Water	0.20	3.5	1.0	2.3
D_2O	0.30	2.8	1.5	2.5
THF	0.80	1.8	4.0	2.0
Propylene carbonate	1.20	0.88	6.0	2.2

Table 7: Electron-transfer rate constants, longitudinal relaxation times, and relative damping ratios for an illustrative five-solvent series. This set spans multiple solvent classes. χ_{rel} (τ_L-only) uses $\chi_A/\chi_B = \tau_L^A/\tau_L^B$. ^{a}Corrected values apply $\chi_A/\chi_B = (\Omega_{s,A}^2/\Omega_{s,B}^2) \times (\tau_L^A/\tau_L^B)$ using Debye-Drude bath frequencies Ω_s extracted from ultrafast solvation dynamics (Horng et al. 1995; Maroncelli 1993): MeCN 100, H_2O 150, D_2O 130, THF 70, PC 60 cm^{-1} (approximate; these are model-specific fitting parameters distinct from far-IR librational peaks). A homologous same-class solvent series eliminates this uncertainty.

The corrected relative χ values compress the series considerably compared to the τ_L-only estimate: the spread reduces from 6× to roughly 2.5×, and the rank order changes (water becomes more overdamped than THF due to its higher bath frequency). The qualitative conclusion, that the series is consistent with sampling the high-friction flank, survives the correction; the location and magnitude of the predicted turnover shift modestly. Because the Debye-Drude Ω_s parameters are model-specific and not universally tabulated, a quantitative Route B test requires either a same-class homologous series or independently fitted Ω_s values from solvation dynamics measurements.

The product $k_{\text{et}} \times \tau_L$ takes values 0.10, 0.07, 0.08, 0.14, and 0.11 across the series. These values vary by roughly a factor of two, indicating that while solvent-dynamical control is the dominant contribution, some variation in thermodynamic parameters (λ, ΔG°) across solvents is likely present. Under the Route B model ($\chi \propto \tau_L$), the observed monotonic decrease in k_{et} with increasing τ_L is *consistent with* sampling the high-friction (spatial-diffusion) side of the Kramers turnover: larger τ_L means larger χ and slower rate, matching the spatial-diffusion limit. Because Route B is relative and lacks an absolute χ anchor, the flank assignment remains provisional until either a full turnover scan or an independently calibrated absolute χ value is available. Using Eq. (4), the relative damping spans a factor of $1.20/0.20 = 6\times$ across the series, with the near-critical optimum predicted to lie at a τ_L shorter than any solvent measured. The reaction-dynamical prediction is that solvent environments with shorter effective relaxation times than those sampled here should eventually enter the underdamped flank and show a rate decrease, with the maximum between the two regimes. This constitutes a falsifiable prediction of the framework.

Proton-coupled electron transfer: mechanism switching via KIE and relative χ

A ruthenium–tyrosine PCET complex [10,11,12] shows pH-independent rate in water but pH-dependent rate in THF. The question is whether the mechanism shift is dynamical or thermodynamic. Measuring KIE (k_H/k_D) and k_{obs} across four solvents of increasing viscosity at fixed T, λ, and ΔG° allows a Route B diagnosis without requiring knowledge of m_{eff}.

For the proton coordinate, H → D substitution increases m_{eff} by a factor of two and therefore *lowers* Ω_{PT} by $\sqrt{2}$ (since $\Omega \propto m_{\text{eff}}^{-1/2}$), shifting:

$$\chi_{\text{eff}}^D = \sqrt{2}\,\chi_{\text{eff}}^H \approx 1.41\,\chi_{\text{eff}}^H \quad \text{(proton-dominated limit)}$$

This relation holds when the proton mass dominates the effective coordinate mass. In heavily solvent-dressed PCET coordinates, where collective reorganization contributes significant effective mass ($m_{\mathrm{eff}} \gg m_{\mathrm{proton}}$), the actual frequency shift upon deuteration is smaller than $1/\sqrt{2}$ and the χ shift correspondingly smaller than $\sqrt{2}$. The expression $\chi_{\mathrm{eff}}^{D} = \sqrt{2}\,\chi_{\mathrm{eff}}^{H}$ therefore represents an upper bound on the isotope-induced χ shift; the qualitative direction (D pushes the coordinate toward larger χ, i.e. more overdamped) is model-independent.

If $\chi_{\mathrm{eff}}^{H} \approx 1$ in a given solvent, the deuterated system sits at $\chi_{\mathrm{eff}}^{D} \approx 1.41$, which is overdamped and stepwise through spatial-diffusion limitation. The KIE trend across the viscosity series encodes the regime traversal: KIE rises as viscosity increases because the system approaches the near-critical window from below and concerted PCET emerges; KIE reaches a feature (local maximum or plateau) as one or both isotope coordinates cross the near-critical window; KIE falls at still higher viscosity because the H coordinate is overdamped and the D coordinate still more so, and the mechanism returns to stepwise through spatial-diffusion sluggishness rather than tunneling collapse. The precise location of the KIE feature depends on the full rate model including intrinsic isotope effects and tunneling contributions; the present framework predicts the qualitative non-monotonic feature rather than its exact friction value. pH-independence and absence of detectable intermediates at the KIE feature confirm the concerted assignment. The non-monotonic KIE is a dynamical signature of the near-critical window crossing, not a tunneling signature, and is consistent with a fully classical framework.

Experimental support for friction-driven mechanism switching comes from Langford and coworkers [28], who studied an excited-state Ru-complex PCET system. Under standard concentrations the reaction proceeds stepwise; at high acceptor and buffer concentrations (100 mM MQ^{+}, 50 mM phosphate) the mechanism switches to concerted PCET. Pressure measurements serve as the diagnostic, not the forcing lever: the stepwise ET pathway exhibits $\Delta V^{\ddagger} = -14.5 \pm 0.5$ $\mathrm{cm}^3\,\mathrm{mol}^{-1}$, consistent with electrostriction from charge separation, while the CPET pathway gives $\Delta V^{\ddagger} = -2.9 \pm 3.4$ $\mathrm{cm}^3\,\mathrm{mol}^{-1}$, statistically indistinguishable from zero. The near-zero activation volume confirms that simultaneous electron-proton transfer bypasses solvent electrostriction entirely, removing the dominant environmental friction channel. This friction-independence is consistent with the boundary condition $\chi_{\mathrm{eff}} \approx 1$ predicted by the framework, providing an experimental diagnostic compatible with the dynamical interpretation above.

A falsifiable prediction is that any control parameter that systematically shifts the effective environmental friction, including solvent, composition, or pressure where applicable, should produce a non-monotonic KIE feature as the system traverses the near-critical window. Failure of such a feature to track the independently identified dynamical crossover at fixed ΔG° and λ would refute the friction-based mechanism assignment (Test 3, Supplementary §4). Diagnostic signatures distinguishing stepwise from concerted PCET are summarized in Table 8.

Observable	Stepwise	Concerted
KIE (k_H/k_D)	2–7	Variable; interpret alongside pH-independence and absence of intermediates
pH dependence	Rate depends on pH	Rate pH-independent
Intermediates	Detectable	None
Pressure response	Large negative $\Delta V^\ddagger$	Near-zero $\Delta V^\ddagger$

Table 8: Diagnostic signatures for PCET mechanism assignment.

Heterogeneous catalysis: operando χ_{spec} as a screening metric for N_2 activation

For catalytic screening, the operational quantity is the damping ratio along the approach coordinate in the precursor well, because within the PGH turnover picture it is the approach dynamics that determine commitment to the barrier; the saddle-point value is treated separately in Supplementary §3.3 as a transition-state bound.

Operando DRIFTS is available on a series of Fe-based catalysts with varying K-promoter loading. N_2 dissociation is the established rate-determining step for ammonia synthesis on Fe under both industrial and model surface-science conditions [17,18], so the dynamical commitment of the N–N bond coordinate is directly kinetically relevant. The objective is to rationalize the TOF-versus-loading curve and predict the over-promotion threshold.

For systems where DFT electronic friction is available [13,14], Route C gives direct estimates. Fitting operando IR to a Voigt profile and extracting $\Delta\tilde{\nu}_L$ (the Lorentzian component only) gives Route A estimates. The two should not be applied to the same linewidth: Route A applies to IR data, Route C applies to DFT Γ_{NN} values; applying Route A mathematics to DFT linewidths produces incorrect results. Canonical values across the Fe surface series, all via Route C from DFT friction [15,16], are given in Table 9.

System	$\tilde{\nu}$ (cm^{-1})	Γ_{NN} (eV)	χ_{DFT}
N_2 gas phase	2359	$\sim 10^{-5}$	$\sim 10^{-5}$
N_2/Fe(111) unpromoted	∼2100	∼0.05	∼ 0.10
N_2/Fe(110)	∼2000	∼0.30	∼ 0.61
N_2/Fe(111) K-promoted	∼1415	∼0.45	∼ 1.28

All χ_{DFT} are ground-state approach-region estimates (χ_0 at ground-state geometry; see Section 2).

Table 9: Approach-region damping ratio across the Fe surface series via Route C (ground-state geometry). Values increase toward $\chi_0 \approx 1$ with K-promoter loading.

The damping descriptor predicts that TOF should maximize near $\chi_{\text{DFT}} \approx 1$ on a log-linear plot of TOF versus χ [19,20,21]. Systems below this value benefit from additional promoter loading; systems above it are over-promoted and lose activity with further loading. Increasing K from zero to the optimal loading simultaneously decreases $\tilde{\nu}$ (via backdonation) and increases $\Delta\tilde{\nu}_L$ (via d-band coupling), driving χ_{spec}

toward unity and raising TOF. Exceeding the optimal loading pushes χ_{spec} beyond the near-critical window ($\chi_0 > 1.3$), and TOF decreases despite continued promoter addition. This prediction is directly falsifiable as Test 7 of the falsification protocol (Supplementary §4): TOF must maximize at finite K loading and decline thereafter.

The Haber–Bosch process conditions (Fe(111), K promoter, 400°C, 200 atm) were arrived at empirically over decades. Retrospectively, the combined effect of promoter loading, elevated pressure, and temperature correlates with χ_{spec} increasing toward $\mathcal{O}(1)$, simultaneously raising Γ_{cat} via d-band coupling and lowering Ω_{bond} via π^*-backdonation. This correlation is consistent with the dynamical interpretation and does not imply that dynamical optimization was the historical design intent. The path toward ambient nitrogen fixation at lower temperature and pressure requires materials in which the approach-region ratio $\chi_0 \approx 1$ can be achieved without those process conditions.

Recent first-principles calculations on ferroelectric $Cu_n(CrSe_2)_{n+1}$ monolayers demonstrate that reversible polarization switching dynamically decouples hydrogen adsorption and desorption, reducing the Heyrovsky barrier from 0.71 eV to 0.36 eV (Tafel: 1.29 eV to 0.72 eV) while incurring a switching cost of only 0.32 eV [27]. Such cycling between electronically distinct adsorption- and desorption-favorable states illustrates an alternative to static binding optimization and suggests that Γ_{eff} may be tuned not only by static promoter loading but also by dynamic switching between electronically distinct surface states.

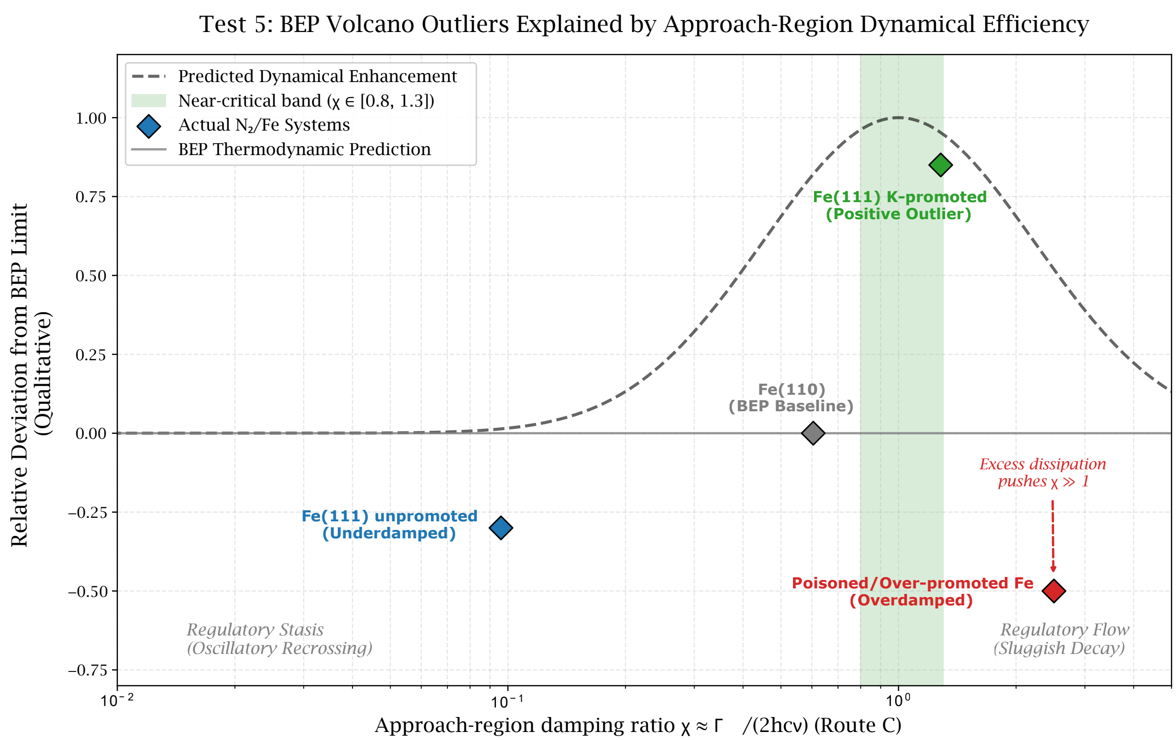

Figure 2: BEP volcano outliers explained by dynamical efficiency (Test 5). Damping ratio $\chi \approx \Gamma_{NN}/(2hc\tilde{\nu})$ (Route C proxy) on the horizontal axis (log scale); deviation from the BEP thermodynamic prediction on the vertical axis. Fe(110) sits at the BEP baseline ($\chi \approx 0.61$; thermodynamics adequate, dynamics partially limiting). Fe(111) unpromoted is a negative outlier ($\chi \approx 0.10$; oscillatory recrossing suppresses activity beyond the thermodynamic expectation). Fe(111) K-promoted is a positive outlier ($\chi \approx 1.28$; near-critical, minimal recrossing). Over-promoted Fe lies in the overdamped regime ($\chi \gg 1.3$; sluggish commitment reduces TOF below the BEP baseline). Dashed curve: predicted dynamical enhancement profile.

For catalytic systems the relevant χ governing turnover is the approach-damping ratio, defined as the value of χ in the precursor well as the coordinate approaches the barrier, not the TS value χ_{TS} at the saddle point itself. In the PGH formulation [6], the turnover maximum is set by the approach-region friction: the ground-state DFT estimates in Table 9 describe this regime, making χ_{GS} the operationally correct quantity for screening. Although χ_{TS} is substantially larger (Supplementary §3.3 brackets $\chi_{\text{TS}} \in [2, 7]$ for K-promoted Fe(111) using the Nosir et al. potential-energy surface [26]), it is the approach dynamics that determine whether the coordinate commits to the barrier or recrosses.

The present framework does not replace d-band theory, BEP scaling relations, or adsorption-energy descriptors; rather, χ_0 provides a strictly reaction-dynamical coordinate that dictates the pre-exponential commitment probability among systems with comparable thermodynamic descriptors, including the anomalous position of K-promoted Fe(111) on the BEP volcano.

Constrained χ-space and the origin of domain-specific optima

The standard-state periodic lattice χ map (Figure 3), where $\chi_{\text{lattice}} = \pi\lambda_{ep}T/(2\Theta_D)$ derives from the Debye-model electron–phonon coupling rate at the bath frequency $\Omega_{\text{bath}} = k_B\Theta_D/\hbar$ [22,23], reveals a structural fact about which systems can access both sides of the $\chi_0 = 1$ boundary. In electron-transfer and PCET systems, both dynamical phases are accessible by varying solvent or composition, and efficiency maxima emerge near the phase boundary for structural reasons. Transition-metal surfaces occupy a more restricted region of χ_0-space: electronic friction at the Fermi level is typically large relative to gas-phase or weakly coupled molecular systems, so well-promoted surfaces tend toward the overdamped side. However, as the Fe series demonstrates ($\chi_0 \approx 0.10$ for unpromoted Fe(111), ≈ 0.61 for Fe(110), ≈ 1.28 for K-promoted Fe(111)), the accessible χ_0-range of a given surface family can span from underdamped through near-critical. Catalytic optimization therefore operates within the accessible band for the specific surface, not in a universally overdamped regime.

This produces a coherent classification (Table 10). The separatrix $\chi_0 = 1$ is a generator-level boundary common to all domains; the accessible χ_0-range is substrate-determined. When a system is structurally confined to one dynamical phase, optimization reorganizes internally to that phase.

Domain	Accessible χ-space	Location of optimum
ET / PCET, solvent dynamics	Both phases	Near separatrix ($\chi_0 \approx 0.8$–1.2)
Metallic surface catalysis	Substrate-dependent; spans under- to near-critical depending on surface and promoter	Near best accessible commitment window; optimized relative to the specific surface family
Wide-gap insulators, noble gases	Underdamped only	Dissipation is not the kinetic bottleneck

Table 10: Accessible χ-space and optimum location by domain.

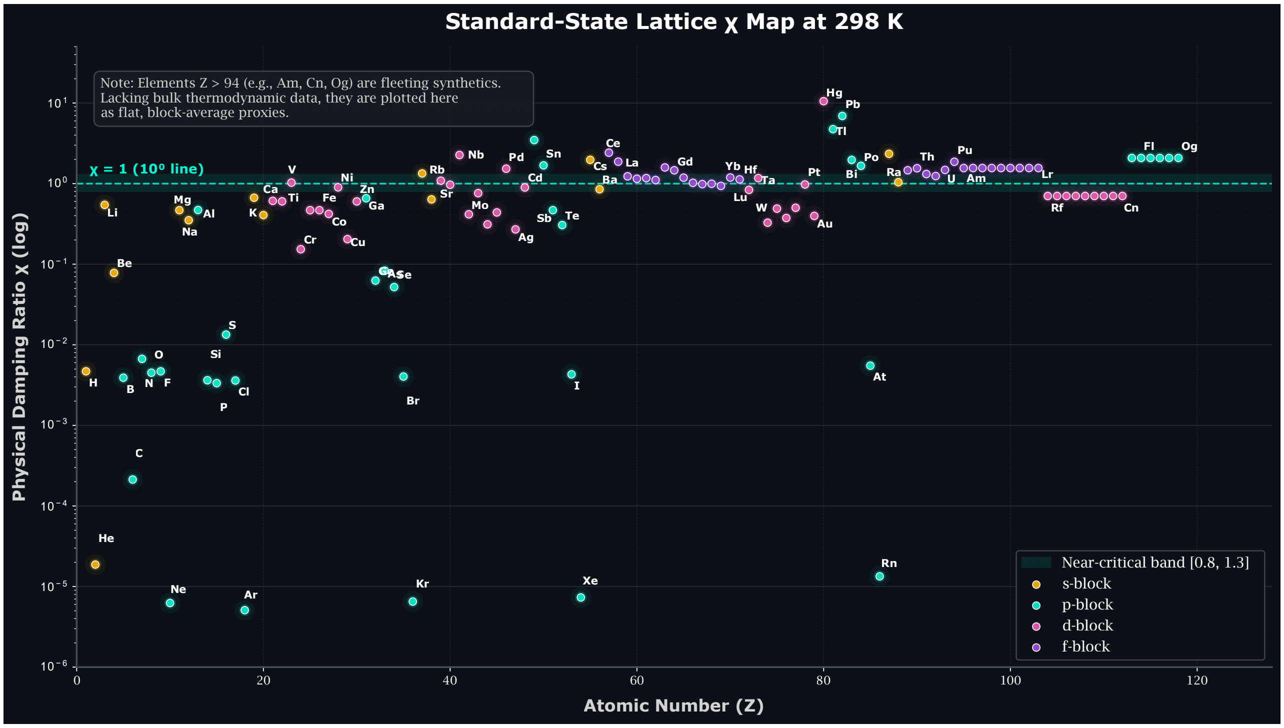

Figure 3: Standard-state lattice χ map at 298 K. $\chi_{\text{lattice}} = \pi\lambda_{ep}T/(2\Theta_D)$ is the Debye-model electron-phonon damping ratio evaluated at the bulk phonon bath frequency. χ_{lattice} is not χ_0; it is a substrate-damping proxy describing how strongly the phonon bath couples to an electronic reactive coordinate at the Fermi level, and therefore which region of χ-space a material inherits as its operating environment. Effective catalytic transition metals span $\chi \approx 0.3$–1.5 at 298 K, straddling the near-critical band; at reaction temperatures (400–700 °C) the active d-block metals shift into $\chi \in [0.8, 1.3]$ as $\chi_{\text{lattice}} \propto T$. Wide-gap insulators and noble gases occupy the underdamped floor; heavy-atom metals (Hg, Pb, Tl) are strongly overdamped. The distribution emerges entirely from independently tabulated λ_{ep} and Θ_D (McMillan 1968 [22]; Allen–Dynes 1975 [23]) and is not imposed by parameter selection. Standard allotrope at STP; elements $Z > 94$ use block-average proxies.

5 Validity Boundaries: When the Framework Does Not Apply

This description applies to classical, thermally activated barrier crossing in a near-Markovian frictional environment. Table 11 gives the conditions required for valid use and those that exclude the framework.

Framework applies	**Framework does not apply**
Thermally activated, over-barrier reaction ($k_BT \gtrsim \hbar\Omega$)	Tunneling-dominated: KIE $>\sim 7$, $T \ll \hbar\Omega/k_B$
Near-Markovian bath ($\Omega\tau_{\text{mem}} \lesssim 1$)	Strongly non-Markovian bath ($\Omega\tau_{\text{mem}} \gg 1$)
Identified dominant reactive coordinate, harmonic near TS	Strongly anharmonic or multi-mode entangled reactive coordinate
Near-adiabatic: moderate electronic coupling H_{ab}	Strictly nonadiabatic: $H_{ab} \to 0$ (friction regime never reached)
Lorentzian fraction $> 50\%$ in IR (Route A)	Inhomogeneous-dominated spectra; raw FWHM at high temperature

Table 11: Validity conditions.

The tunneling boundary is the most commonly encountered exclusion condition. A KIE greater than approximately 7 for H/D indicates substantial hydrogen tunneling contribution; the χ description covers the classical component only, and instanton or open-quantum-systems methods are required for the tunneling component. For systems near this boundary, the classical and tunneling contributions can be separated by their distinct temperature dependences before applying this analysis to the classical component.

When $\tau_{\text{mem}} \sim 1/\Omega_{\text{TS}}$, the Grote–Hynes [7] frequency-dependent friction applies. The dissipative (real) component of the Debye-type memory kernel gives $\Gamma_{\text{eff}}(\Omega) \approx \gamma_0(1 + (\Omega\tau_{\text{mem}})^2)^{-1}$; the full response magnitude gives $|\hat{\gamma}(\Omega)| \approx \gamma_0(1 + (\Omega\tau_{\text{mem}})^2)^{-1/2}$. The dissipative form is the physically appropriate choice for frictional damping; both expressions reduce to γ_0 in the Markovian limit. The $\chi_0 = 1$ boundary broadens into a band under either convention, but the boundary persists. For $\Omega\tau_{\text{mem}} \lesssim 0.3$, the Markovian approximation is adequate. For $\Omega\tau_{\text{mem}} \sim 1$, the target window should be widened to $[0.5, 1.5]$ to account for memory-induced broadening.

6 Summary of Estimator Equations and Decision Criteria

Estimator routes and regime thresholds are collected in Table 12 for reference. A supplementary cross-domain consistency check is described in Supplementary §4. Because that analysis depends on genuinely independent optimal systems drawn from different chemical families, it is presented there as a supporting consistency test rather than as a primary validation pillar of the reaction-dynamical argument.

Estimator	Equation	Validity condition
Route A (spectroscopic)	$\chi_{\text{spec}} = \Delta\tilde{\nu}_L/(2\tilde{\nu})$	Lorentzian fraction > 50%; coverage-independent linewidth
Route B (solvent relaxation, relative)	$\chi_{\text{ET}}^{(A)}/\chi_{\text{ET}}^{(B)} = \tau_L^{(A)}/\tau_L^{(B)}$	Preferred; larger $\tau_L \Rightarrow$ larger χ. Absolute: bracket over C_{solv} and m_{eff} ranges
Route C (DFT friction)	$\chi_{\text{DFT}} = (\Gamma_{\text{int}} + \Gamma_{\text{cat}})/(2hc\tilde{\nu})$; Γ_{cat} in eV	Mean-field DFT valid; Γ_{cat} is energy width (N-A), not friction coefficient; treat as lower bound on χ_{TS}
Regime thresholds: $\chi < 0.8$ underdamped; 0.8–1.3 near-critical (optimal); $\chi > 1.3$ overdamped		
Framework not applicable: KIE > 7; Gaussian > Lorentzian in IR; $\Omega\tau_{\text{mem}} > 1$; $H_{ab} \to 0$		

Table 12: Summary of estimator equations, validity conditions, and regime thresholds.

More broadly, the results place electron transfer, PCET, and surface reaction turnover within a common barrier-crossing language that is native to chemical physics while remaining directly testable in chemically specific systems.

Author contributions

N. Christensen: Conceptualization; Methodology; Formal analysis; Investigation; Writing – original draft; Writing – review and editing; Visualization.

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Conflicts of interest

There are no conflicts to declare.

Data availability

All numerical data and Python code supporting this work are deposited at Zenodo (10.5281/zenodo.17891562) and GitHub (`https://github.com/SymCUniverse/Chemistry`). The computational implementation is also provided in the Supplementary Information. References cited in the Supplementary Information are included in the reference list of this article.

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Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Claude (Anthropic), GPT-4 (OpenAI), DeepSeek (DeepSeek AI), Gemini (Google), and Microsoft Copilot in order to assist with language refinement and cross-check derivational consistency. After using these tools, the author reviewed and edited all content as needed and takes full responsibility for the content of the published article.

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